Learn C++ Programming -Beginner to Advance- Deep Dive in C++

Section 5:C++ Basics

Lecture 24. Why Data Types ?

1. Introduction

A program is made up of two main ingredients:

Data
Instructions / Operations

Instructions or operations are performed on data to get computational results.

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2. Program Execution in Memory

When a program runs, memory is divided into different sections:

Main Memory

- **Heap**
- **Stack**
- **Code Section**

The code section contains instructions.

The stack contains data, such as variables.

The heap is another memory area used during program execution.

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3. Data and Instructions

Data is used by the program.

Instructions are used to perform operations on that data.

Example:

```
a = 10
b = 5
c = a + b
```

Here:

10
5

are stored as data.

The operation:

```
a + b
```

is performed on the data.

The result is stored in:

c

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4. Data in Daily Life

Data is used in daily life in many forms.

Examples:

Bill amount
Price of an item
Roll number
Marks
Door number
Name of a person
Name of a place

These are all examples of data.

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5. Types of Data

Data can be categorized mainly into:

Numeric data
Character data

Sometimes data can be a mixture of numbers and alphabets.

Example:

125-A

This contains both a number and an alphabet.

This can be called:

Alpha-numeric data

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6. Numeric Data

Numeric data means numbers or figures.

Examples:

12
25
15.35

Numeric data can be divided into two types:

Integer
Floating point

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7. Integer Data

Integer data means numbers without a decimal point.

Examples:

12
25

These are integer values.

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8. Floating Point Data

Floating point data means numbers with a decimal point.

Example:

15.35

Floating point means decimal.

The method of representing decimal numbers is called floating point.

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9. Integer and Floating Point Difference

In daily life, we usually call both integer and decimal values numbers.

But in computers, they are treated differently.

Numbers are represented in binary form.

Decimal point numbers need extra handling.

So, numbers without decimal points are treated as integer values.

Numbers with decimal points are treated as floating point values.

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10. Character Data

Character data means alphabets, words, sentences, or names.

Examples:

Mark
Anil
Delhi
program

A single letter is called a character.

A collection of characters forms a string.

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11. String

A string is a collection of characters or alphabets that form a single entity.

Examples:

Mark
Anil
Delhi
program

A string may be:

Name of a person
Name of a place
A word
A song

C++ supports both:

Individual characters
Collection of characters

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12. Binary Representation

Numbers are represented in binary form.

Characters are also represented in binary form.

For characters, special codes are used to represent them.

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13. Memory and Bytes

Data has to be stored in memory so that the program can work on it.

Memory is represented in terms of bytes.

One byte is a collection of:

8 bits

A byte stores data in the form of:

0s and 1s

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14. One Byte Storage

In one byte, the minimum value that can be stored is:

0

The maximum value that can be stored is:

255

So, one byte cannot store values bigger than 255.

To store bigger values, more than one byte is required.

Examples:

2 bytes
4 bytes

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15. Size of Data Types

Different data types take different amounts of memory.

Examples:

Integer data
Floating point data
Character data

Each data type has its own memory size.

The number of bytes required depends on the data type.

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16. Variables

A variable is used for storing data.

Example:

```
int a, b, c;
```

Here, a, b, and c are variables.

If integer takes 2 bytes, then each variable gets 2 bytes.

If integer takes 4 bytes, then each variable gets 4 bytes.

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17. Using Variables

Variables are used by their names in a program.

Example:

```
a = 10;
b = 5;
c = a + b;
```

Here:

10 is stored in a
5 is stored in b
a + b is calculated
result is stored in c

In programs, variable names are used instead of memory addresses.

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18. Variable and Data Type

A variable stores data.

The type of data must also be mentioned.

Example:

```
int a;
```

Here, a can store integer type data.

Integer does not allow decimal values.

If decimal values are required, the variable should be declared as float.

Example:

```
float a;
```

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19. Main Idea

Data types are needed because different kinds of data are stored and handled differently.

Examples:

Integer data
Floating point data
Character data
String data

Each type of data has a different purpose and may require a different amount of memory.

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20. Key Points

A program contains data and instructions.

Instructions perform operations on data.

Data is stored in memory during program execution.

The stack contains variables.

The code section contains instructions.

Data can be numeric or character type.

Numeric data can be integer or floating point.

Integer data has no decimal point.

Floating point data has a decimal point.

Character data contains alphabets or symbols.

A string is a collection of characters.

Data is stored in binary form.

Memory is represented in bytes.

One byte has 8 bits.

A variable is used for storing data.

A variable name is used in the program instead of memory addresses.

The data type tells what type of data can be stored in a variable.